r1.high, step 2 (2026-11-02)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals
1	20263016	5000	0.277	−37234.7	−37221.2	0
2	20263017	5000	0.278	−37148.8	−37126.5	0
3	20263018	5000	0.277	−37164.0	−37151.7	0
4	20263019	5000	0.276	−37160.7	−37145.7	0

file
mcmc_r1.high_step2_chain1_2026-10-30.Rdata
mcmc_r1.high_step2_chain2_2026-10-31.Rdata
mcmc_r1.high_step2_chain3_2026-10-31.Rdata
mcmc_r1.high_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

age.0.to.14.hiv.mortality.multiplier.1	trate.3	birth.transmission.risk.0
3.281	2.927	2.905
age.30.to.39.transmission.multiplier.3	age.15.to.24.hiv.mortality.multiplier.1	age.20.to.29.transmission.multiplier.1
2.856	2.768	2.420
hiv.specific.mortality.rates.1	age.50.and.over.transmission.multiplier.2	log.OR.engagement.slope
2.408	2.081	2.052
age.30.to.39.transmission.multiplier.2	female.to.male.multiplier	trate.1
2.022	1.943	1.861
age.25.to.49.hiv.mortality.multiplier.1	suppressed.disengagement.rates	hiv.specific.mortality.rates.2
1.828	1.802	1.791
age.50.and.over.transmission.multiplier.3	age.30.to.39.transmission.multiplier.1	age.40.to.49.transmission.multiplier.2
1.776	1.745	1.698
age.25.to.49.hiv.mortality.multiplier.0	trate.2	male.awareness.multiplier
1.643	1.594	1.591
age.15.to.24.hiv.mortality.multiplier.2	age.15.to.19.transmission.multiplier.3	male.engagement.multiplier
1.567	1.543	1.516
age.20.to.29.transmission.multiplier.3		
1.484		

Merged simset:

location	simulations	total.weight	file
r1.high	1000	1000	simset_r1.high_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000–2025): 27 of 33 targets flagged

data.type	age	sex	years.with.data	pct.years.in.sim.95	median.sim.vs.data	last.year	last.sim.vs.data
hiv.mortality	50 and over	both	26	42	x1.51	2025	x1.41
suppression	0–14	both	11	0	+12.9 pts	2025	+10.2 pts
hiv.mortality	All ages	both	26	27	x1.32	2025	x1.83
hiv.mortality	10–19	both	26	19	x0.79	2025	x0.65
incidence	15–24	both	26	8	x0.85	2025	x0.80
prevalence	15–24	both	26	8	x0.85	2025	x0.82
engagement	0–14	both	11	0	+5.2 pts	2025	+5.8 pts
engagement	15+	female	11	0	+4.9 pts	2025	+6.8 pts
awareness	0–14	both	11	0	−4.6 pts	2025	−4.3 pts
awareness	15+	male	11	0	−4.1 pts	2025	−2.5 pts
suppression	15+	female	11	9	+4.5 pts	2025	+4.7 pts
hiv.mortality	15–24	both	26	35	x0.86	2025	x0.73
prevalence	All ages	both	26	4	x0.93	2025	x1.35
awareness	15+	female	11	0	+2.8 pts	2025	+2.8 pts
engagement	All ages	both	11	0	+2.8 pts	2025	+3.0 pts
engagement	15+	both	11	0	+2.8 pts	2025	+3.0 pts
prevalence	0–14	both	26	15	x1.00	2025	x0.93

r1.high step 2: likelihood

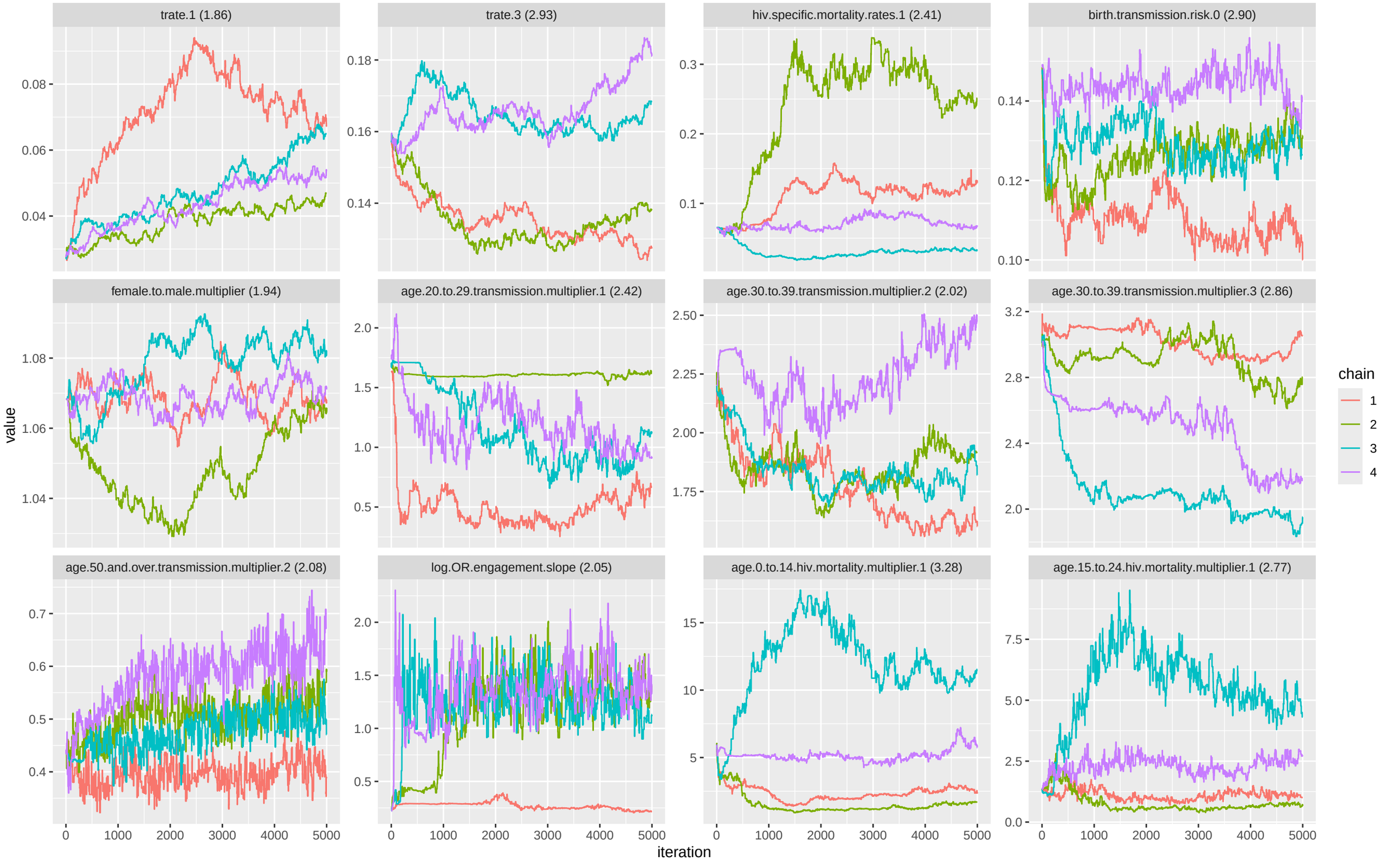
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

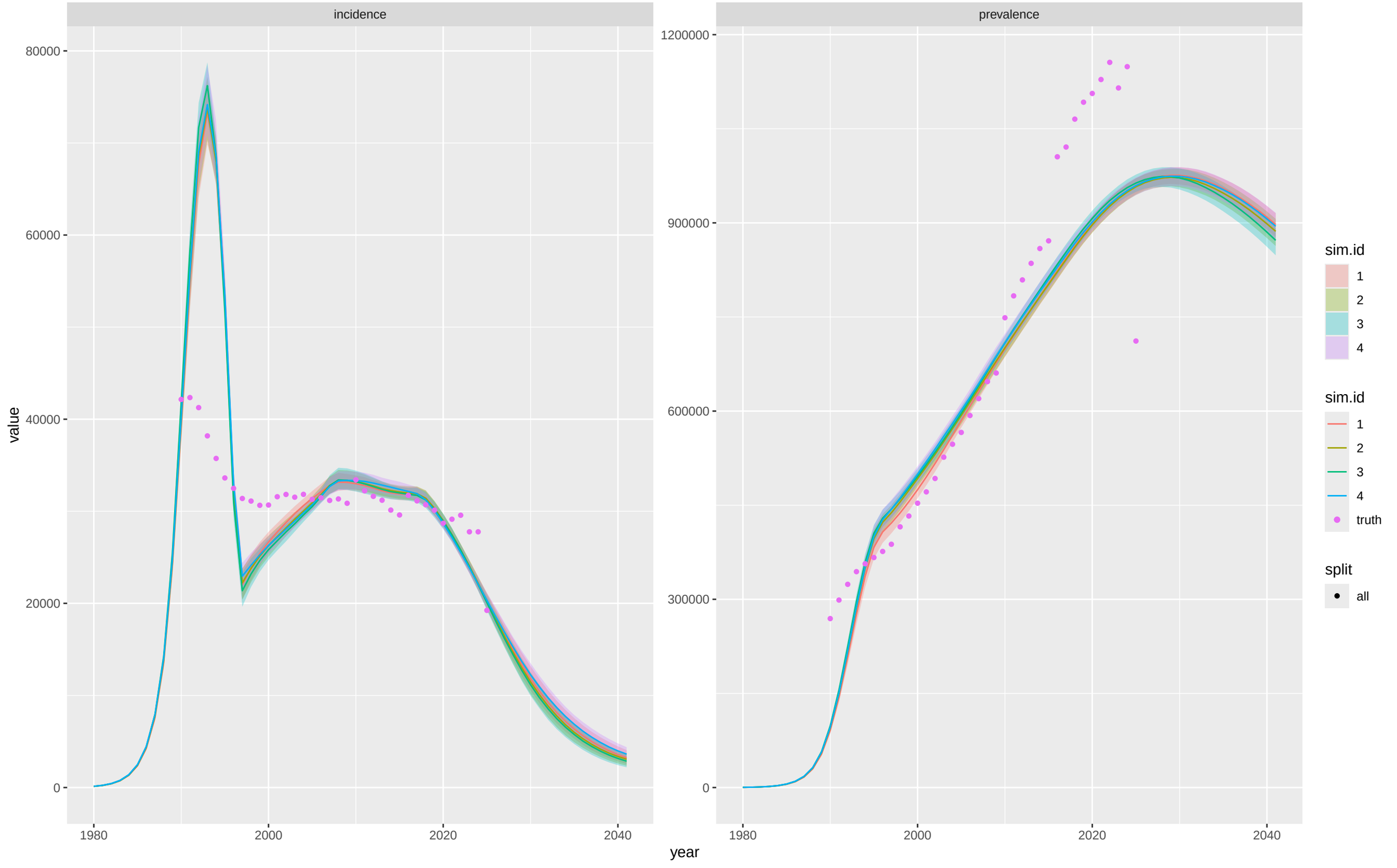
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

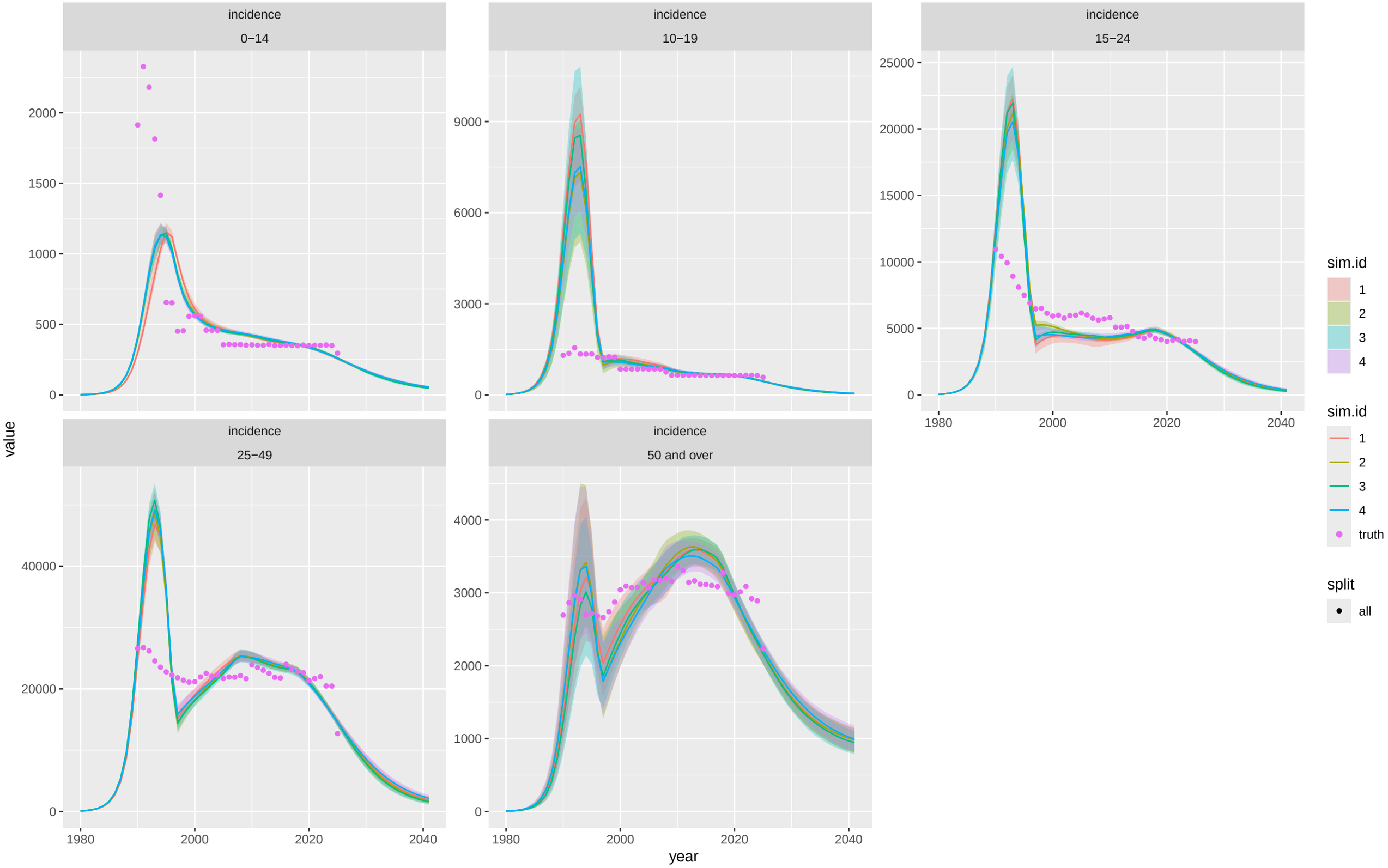
r1.high step 2: trace



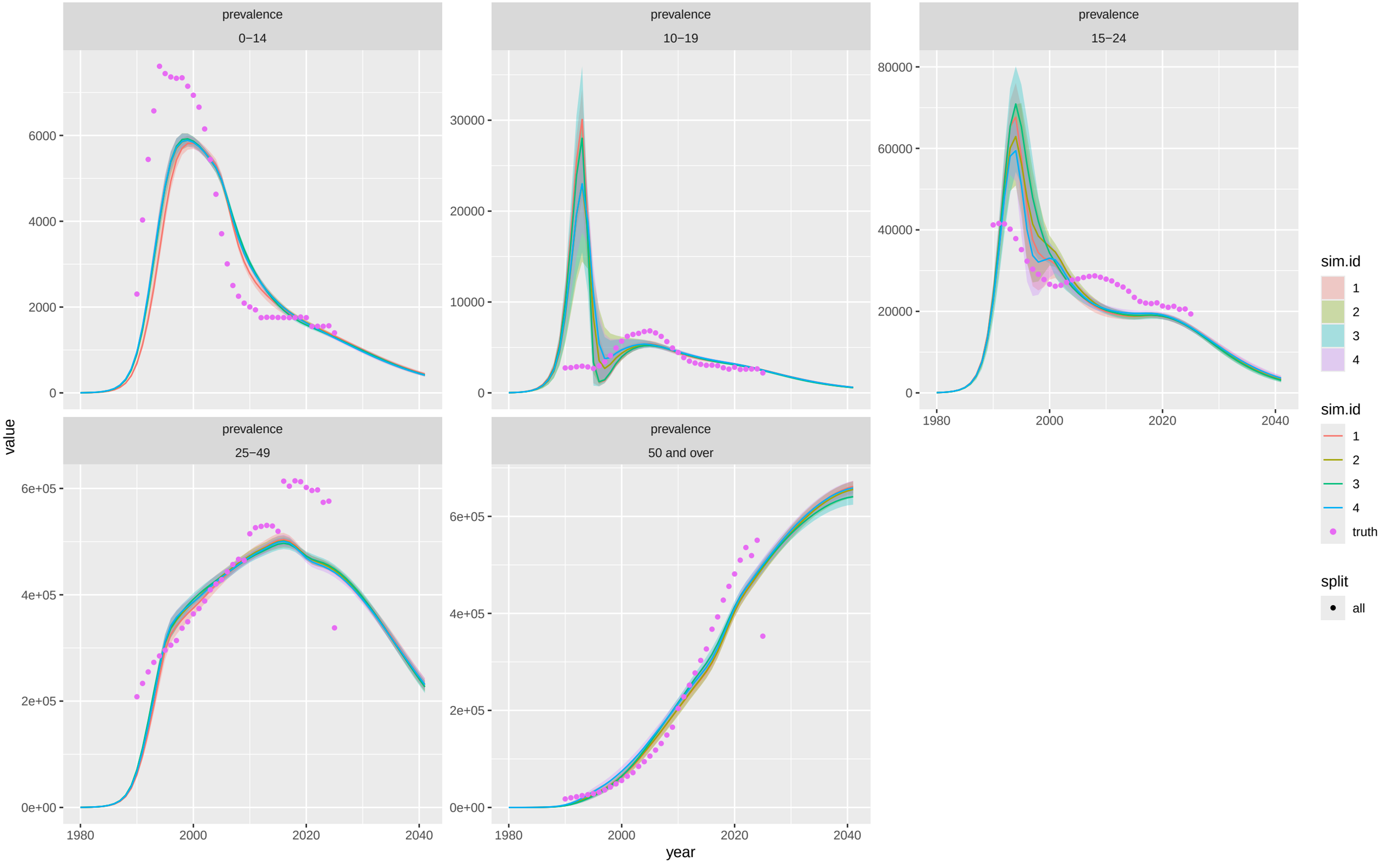
r1.high step 2: fit (one color per chain)



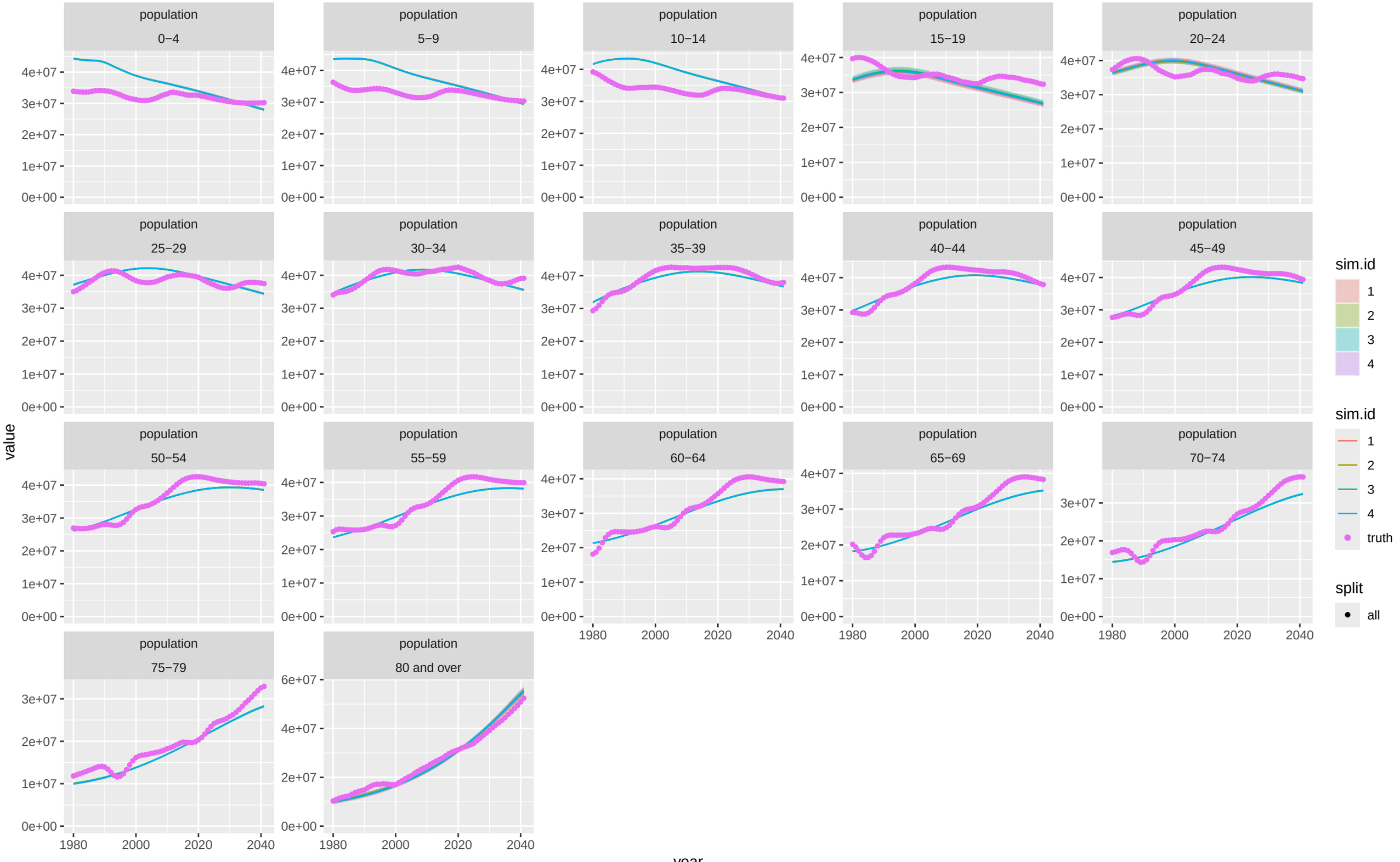
r1.high step 2: incidence.by.age (one color per chain)



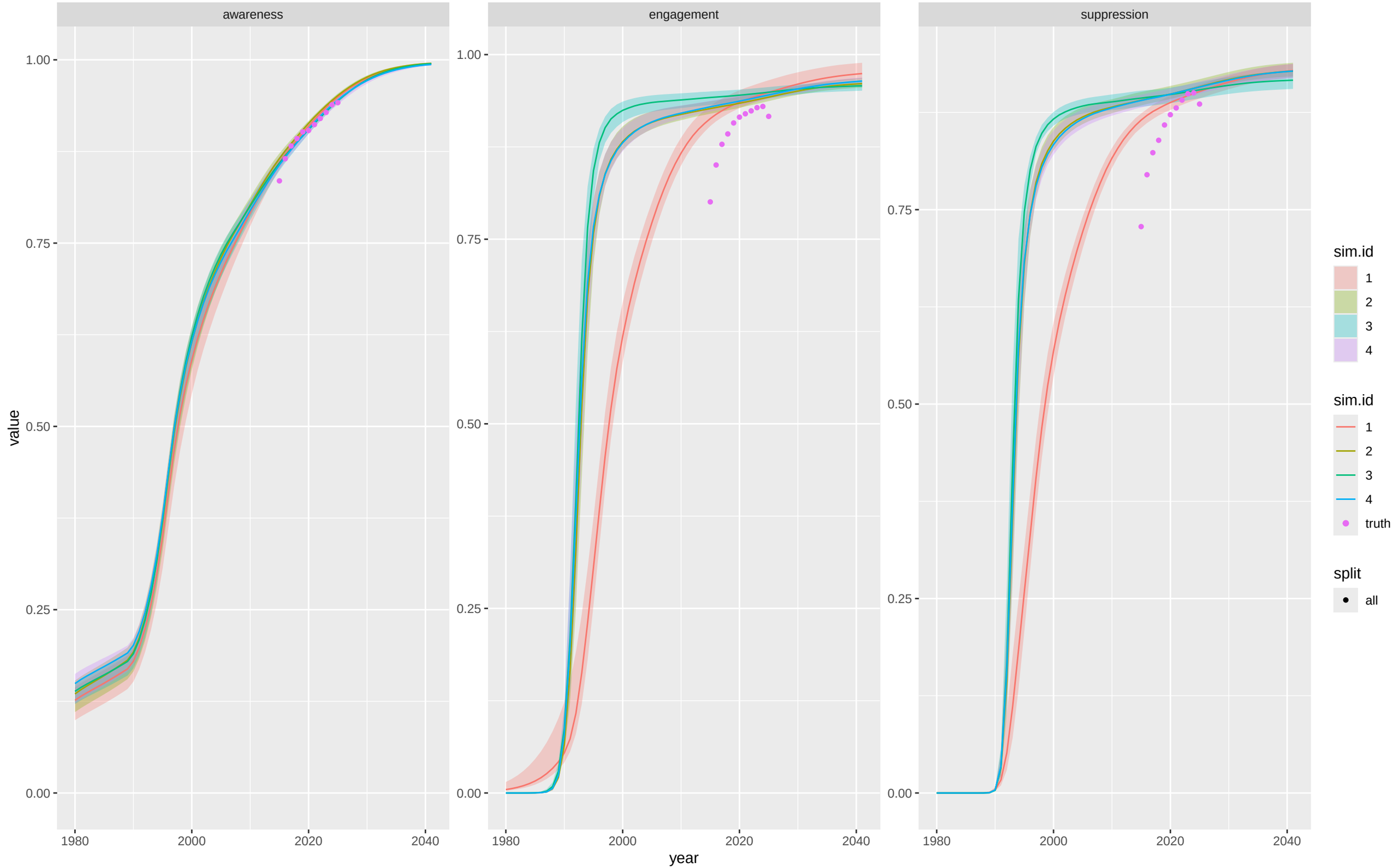
r1.high step 2: prevalence.by.age (one color per chain)



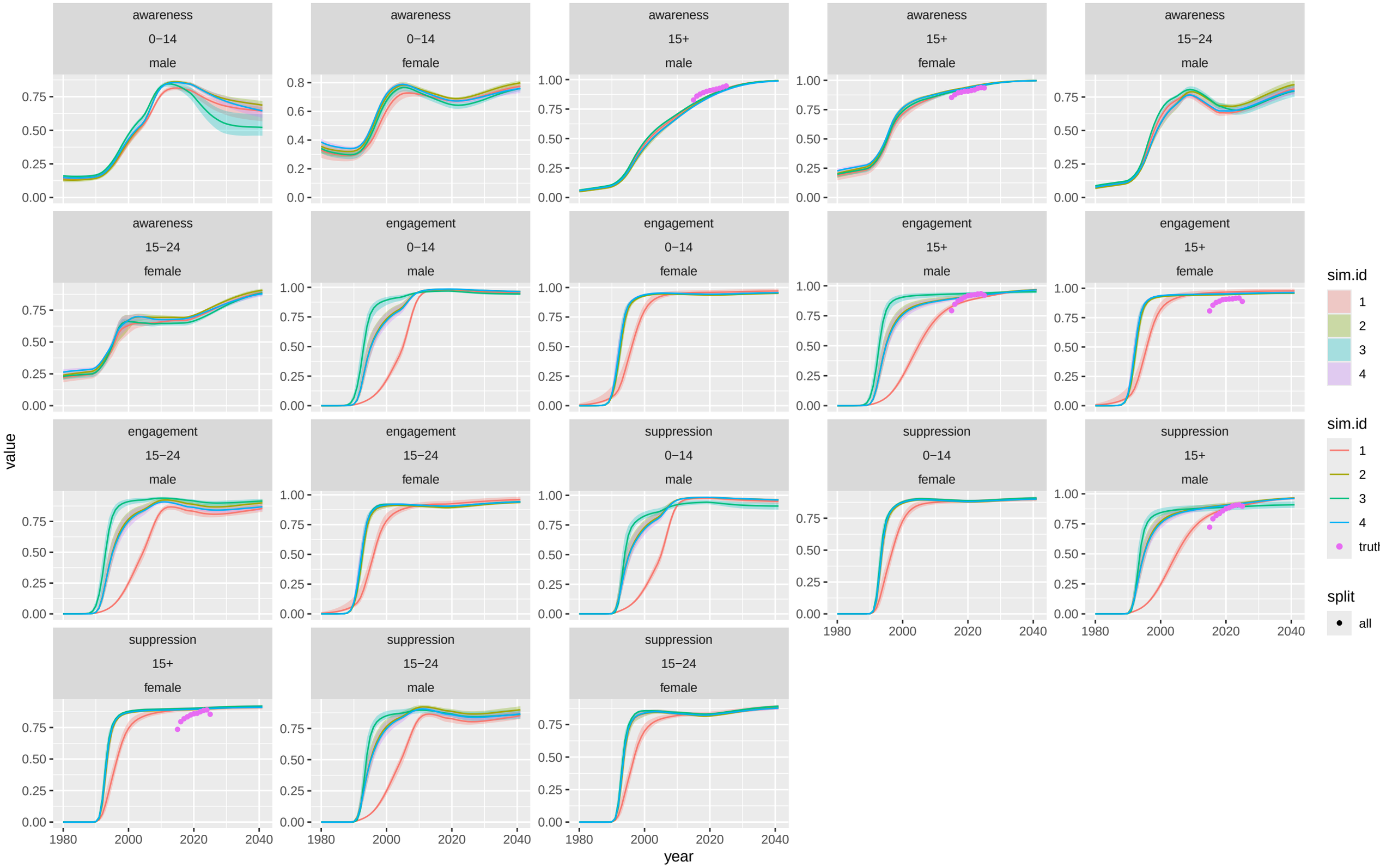
r1.high step 2: population.by.age (one color per chain)



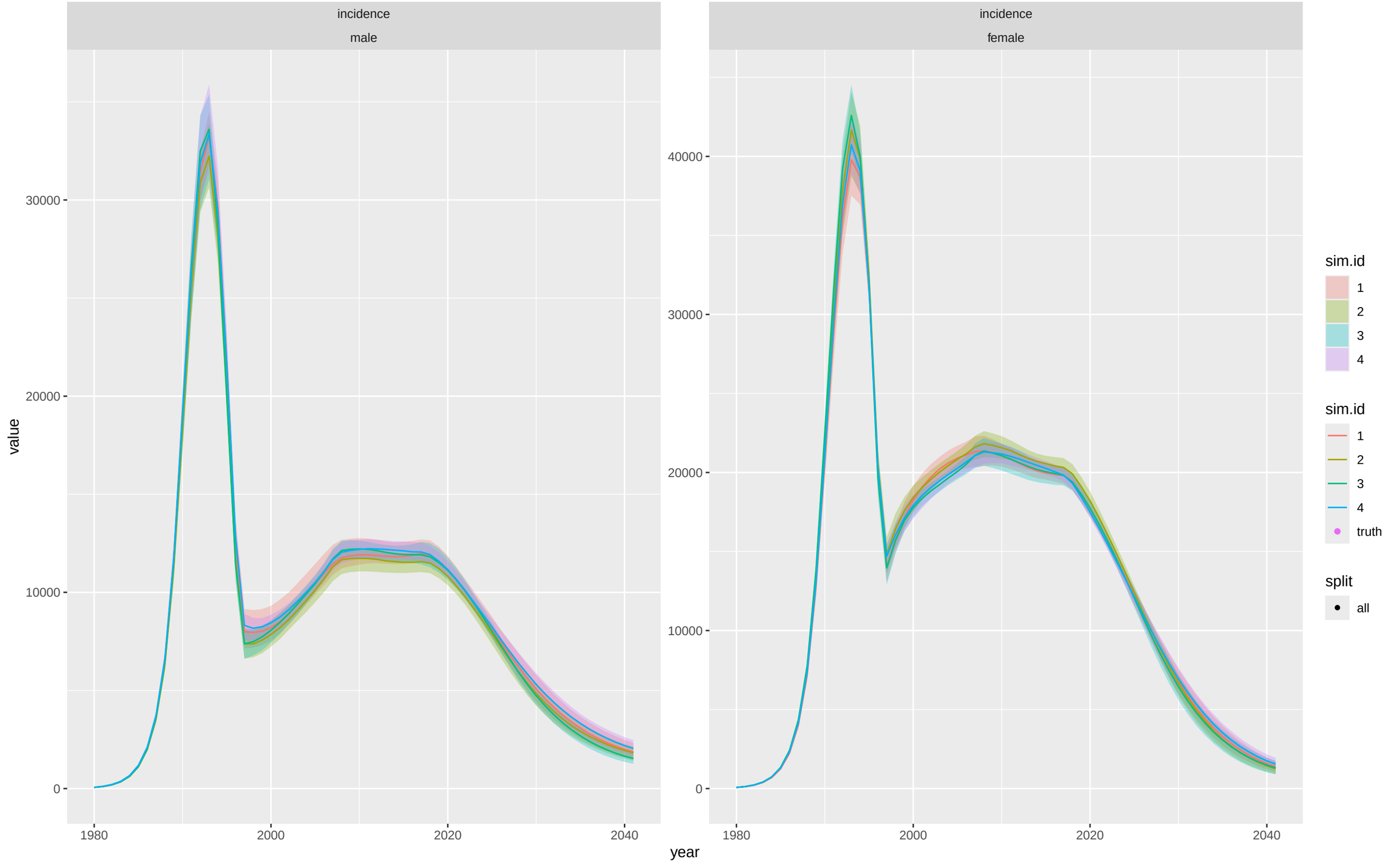
r1.high step 2: cascade (one color per chain)



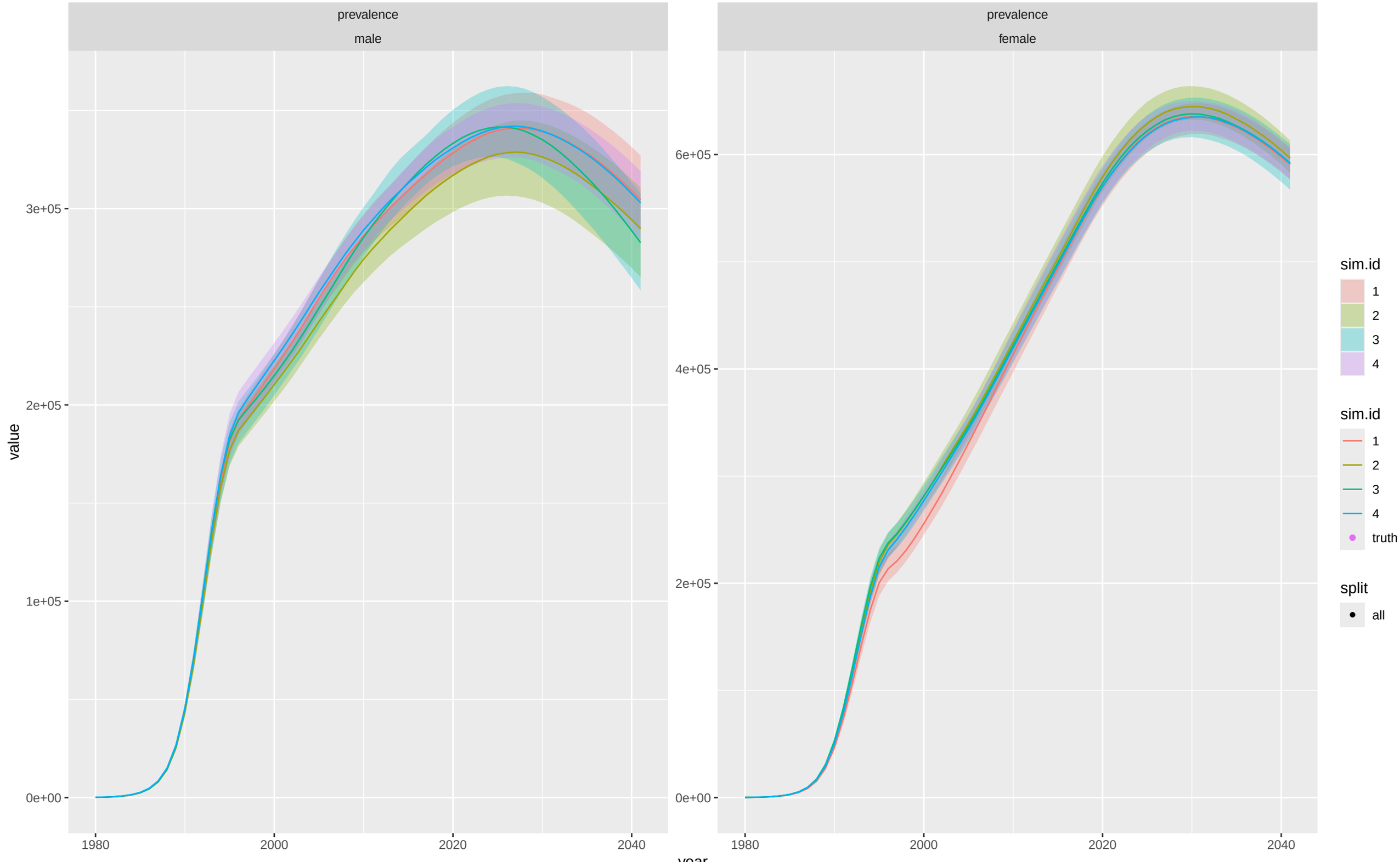
r1.high step 2: cascade.by.age.sex (one color per chain)



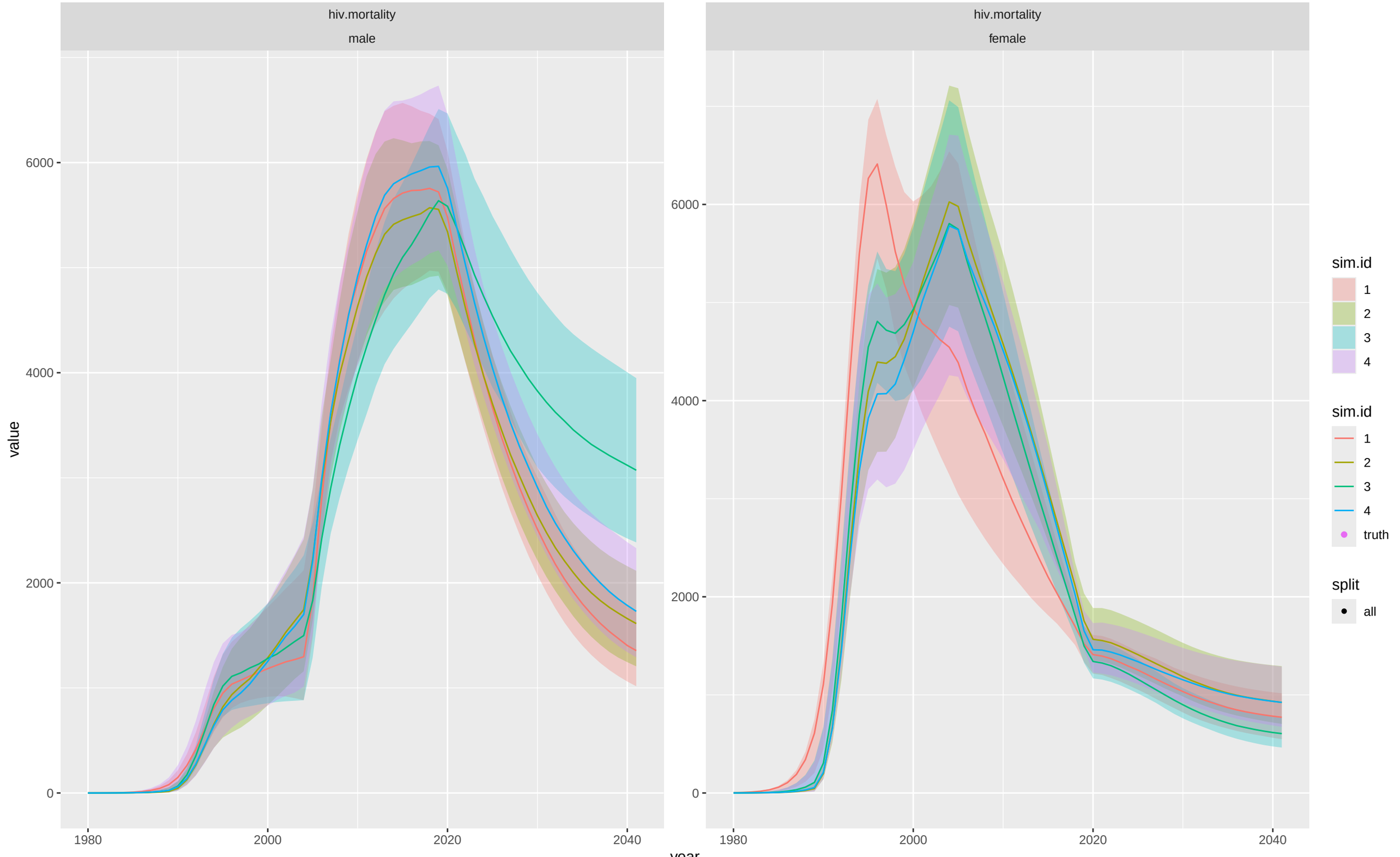
r1.high step 2: incidence.by.sex (one color per chain)



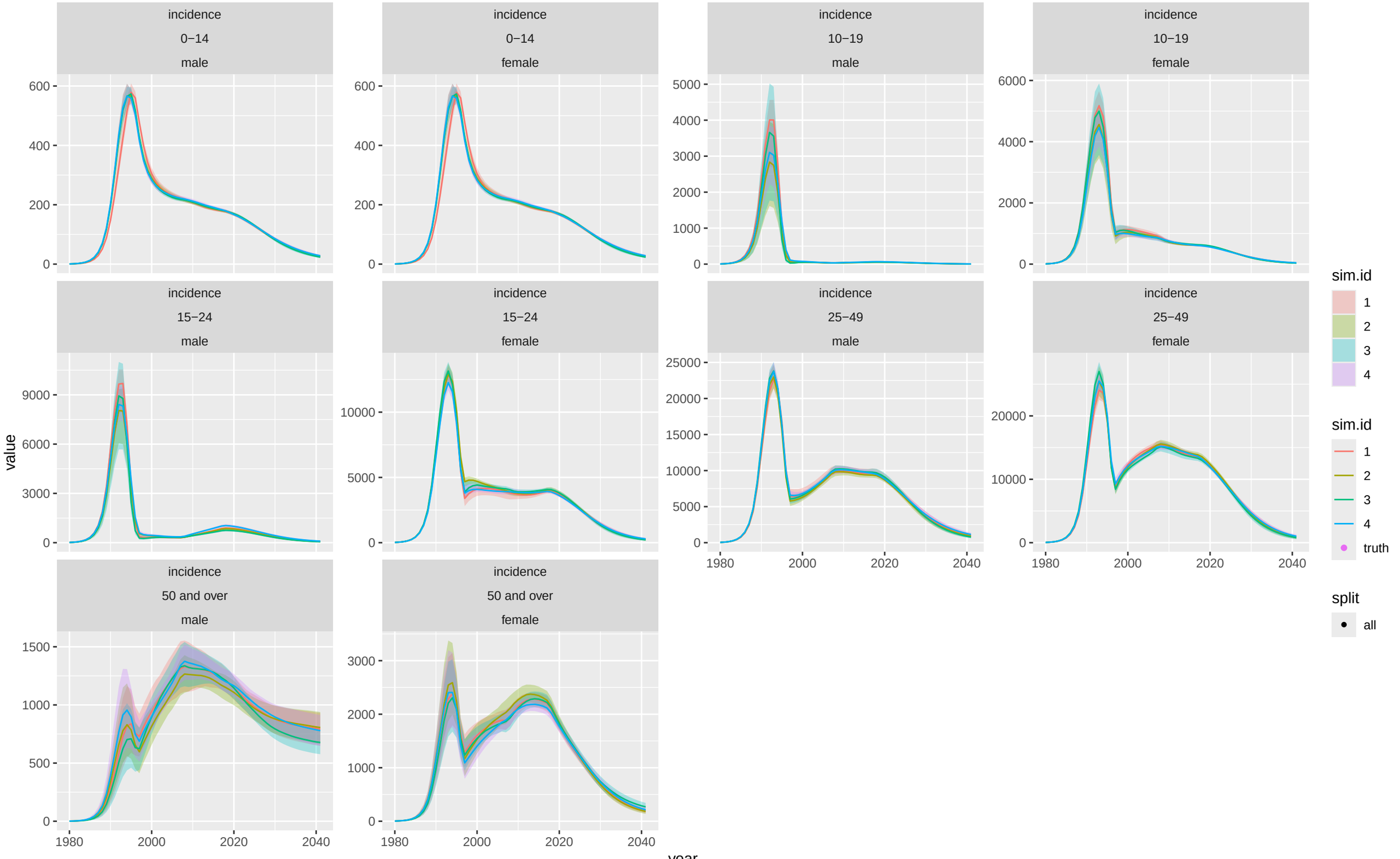
r1.high step 2: prevalence.by.sex (one color per chain)



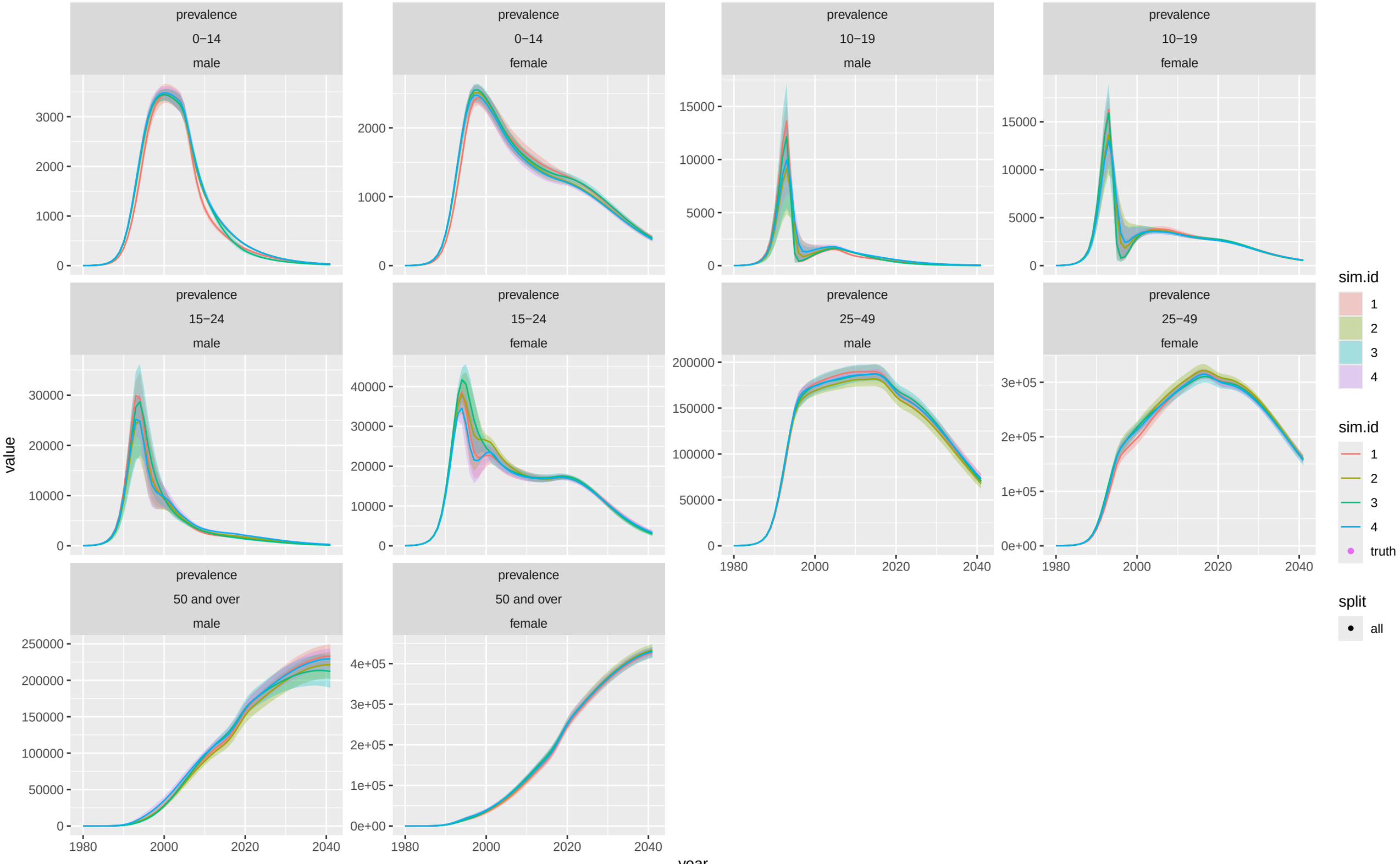
r1.high step 2: hiv.mortality.by.sex (one color per chain)



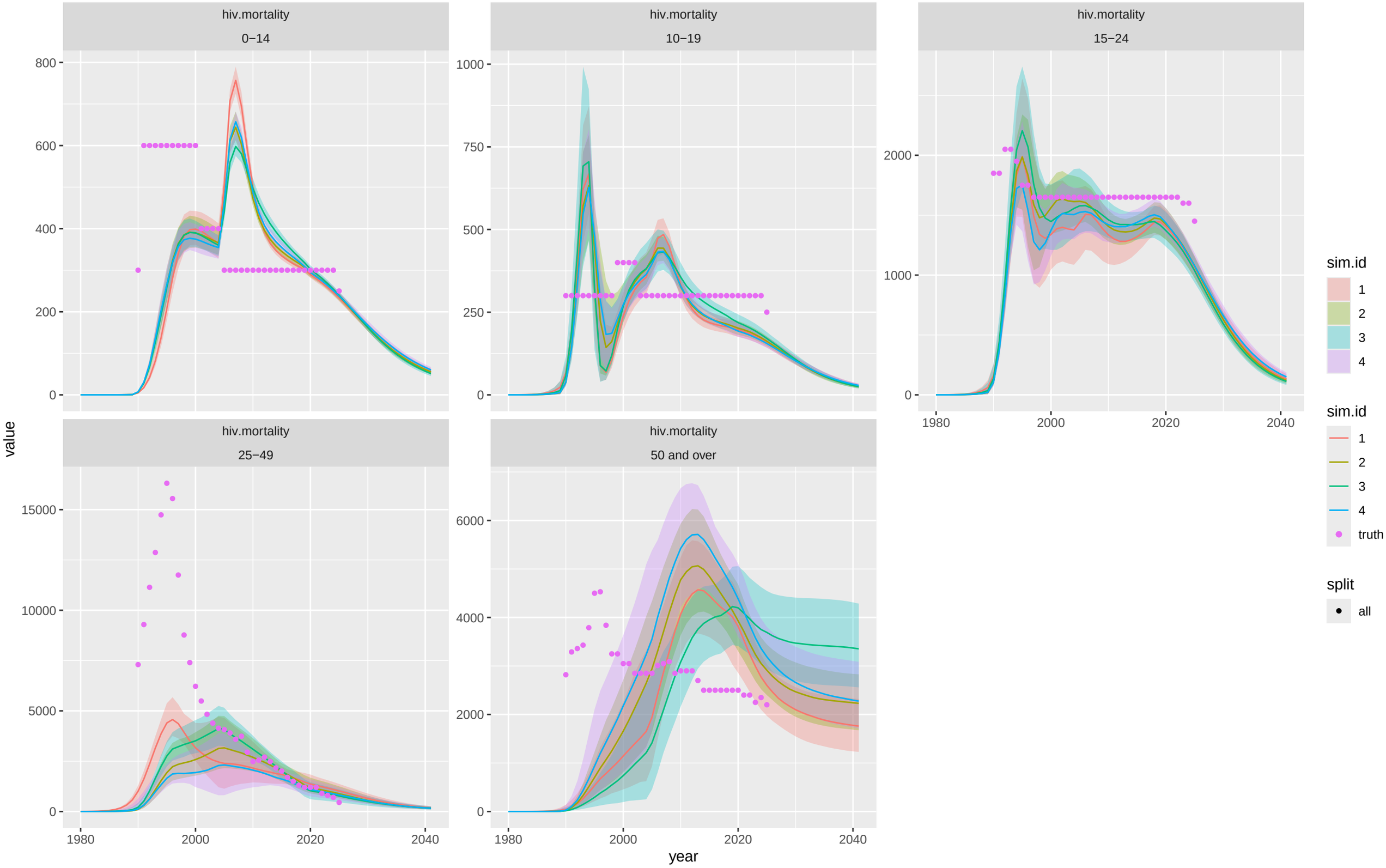
r1.high step 2: incidence.by.age.sex (one color per chain)



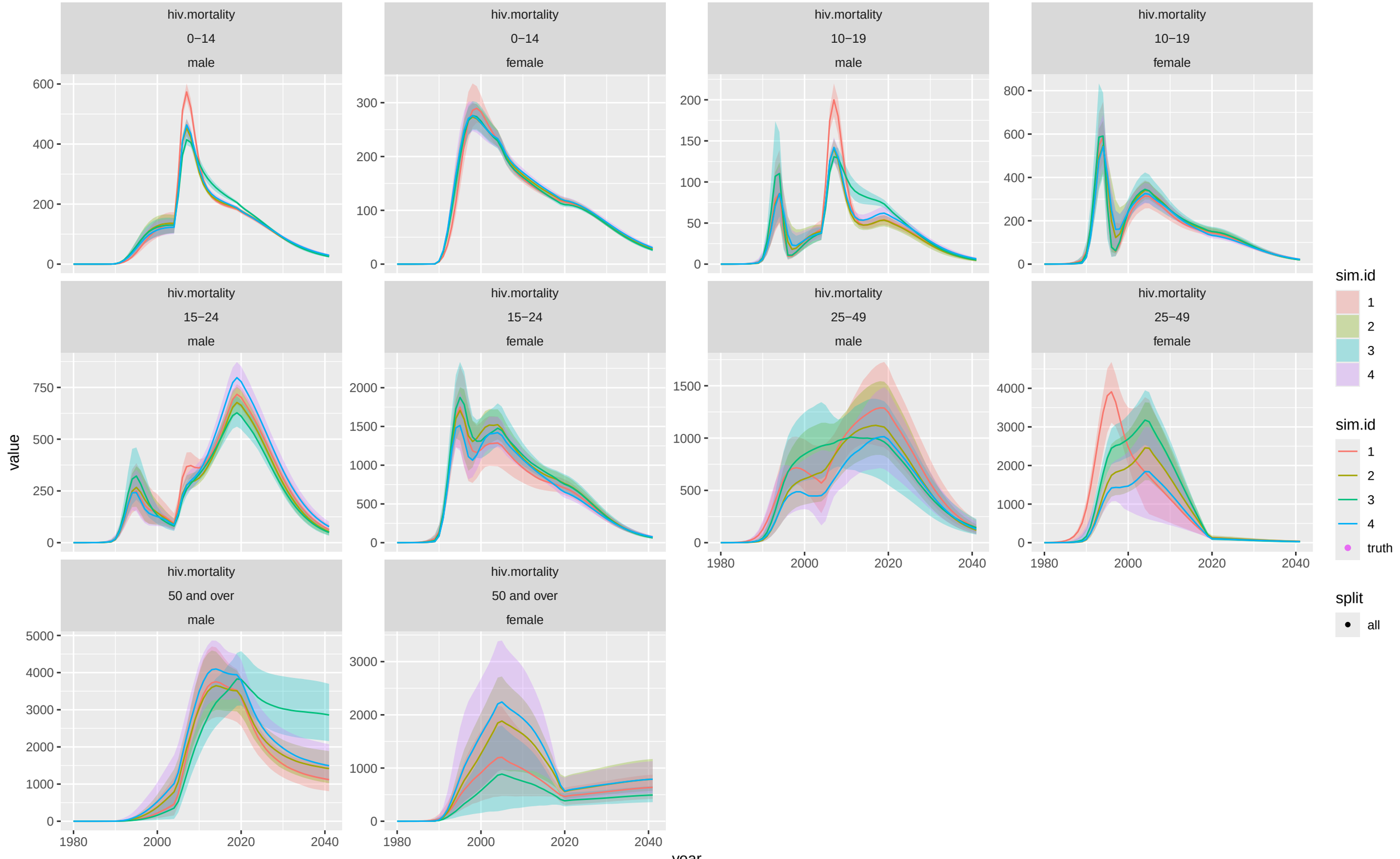
r1.high step 2: prevalence.by.age.sex (one color per chain)



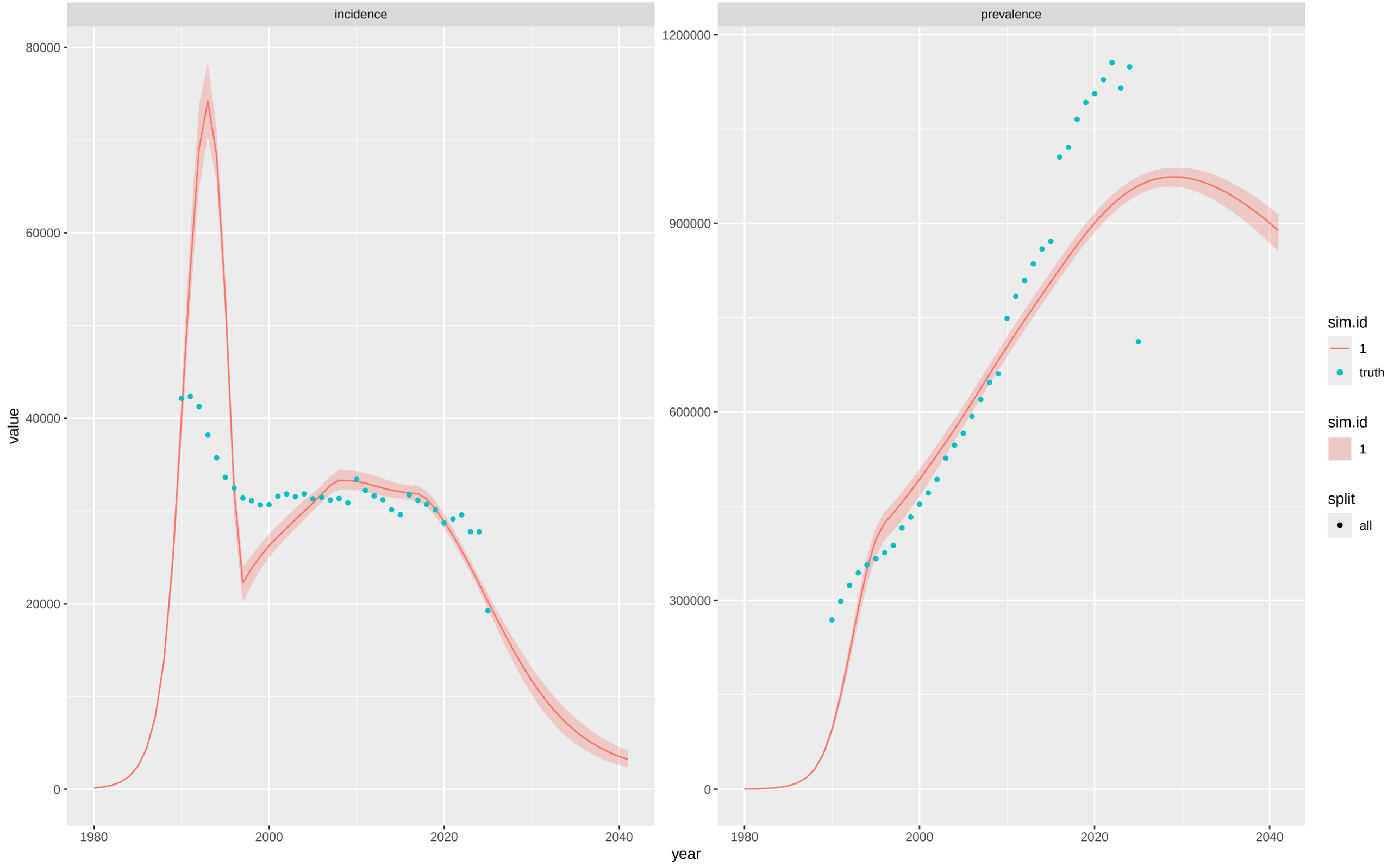
r1.high step 2: hiv.mortality.by.age (one color per chain)



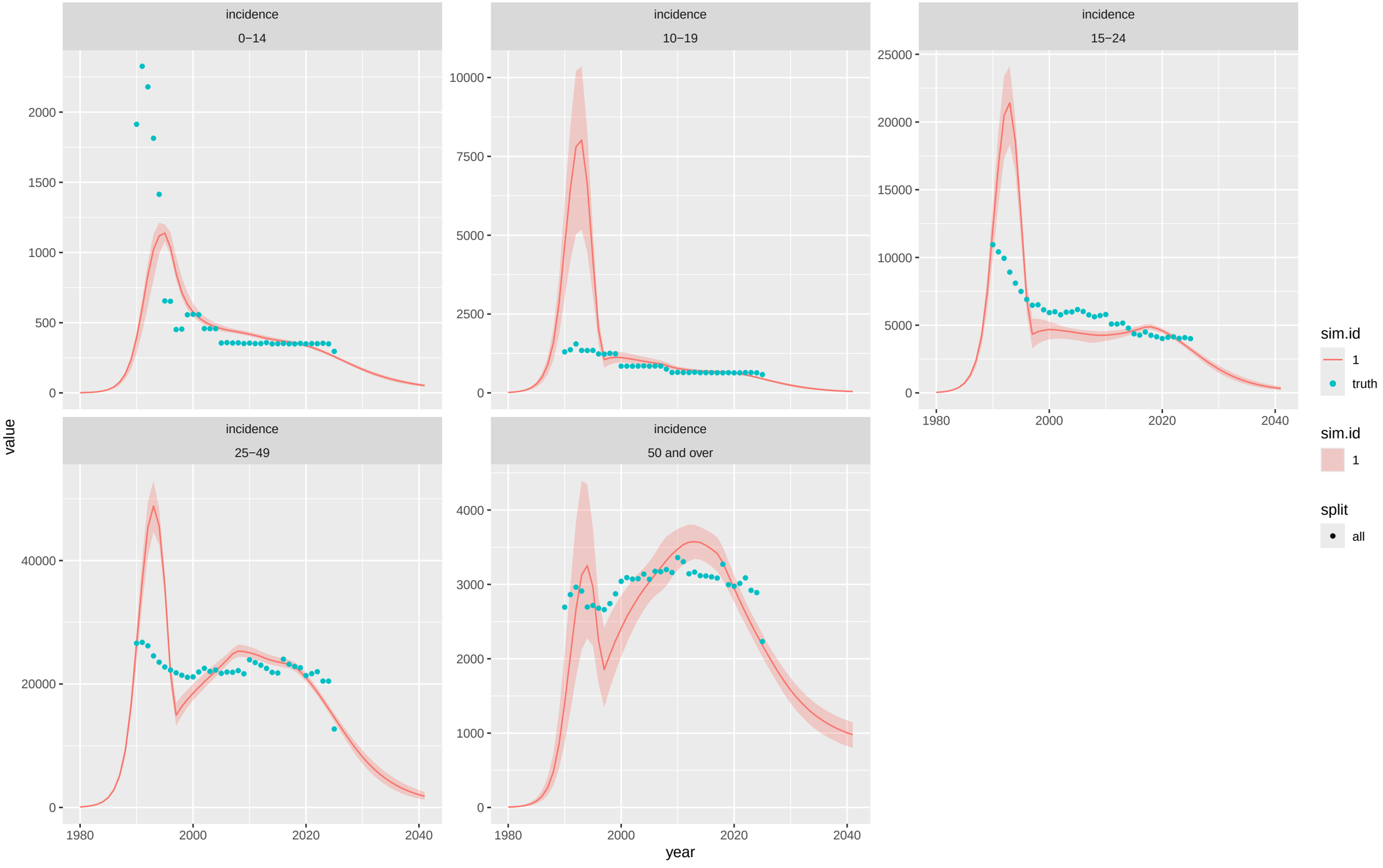
r1.high step 2: hiv.mortality.by.age.sex (one color per chain)



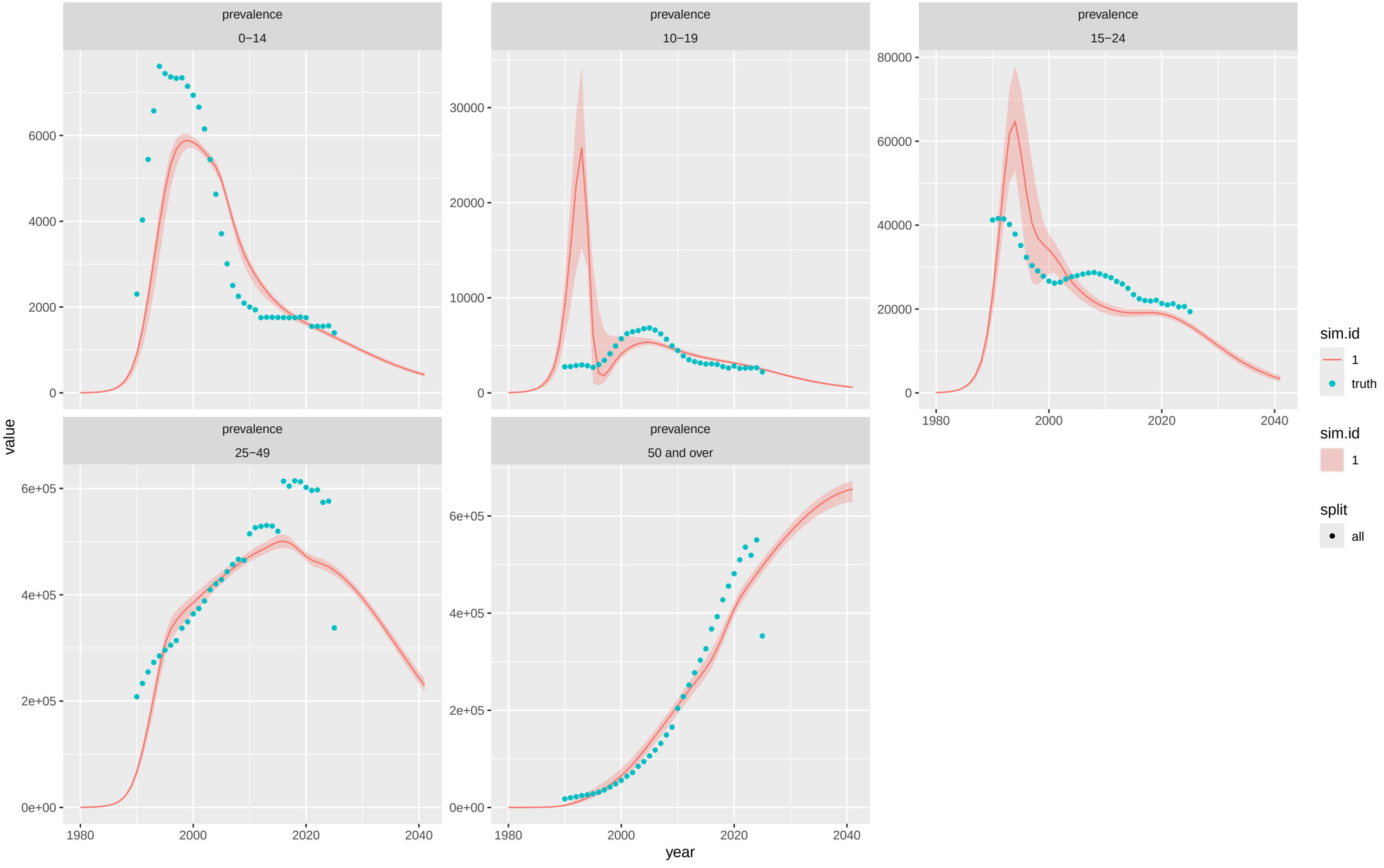
r1.high: fit (merged Step 2 simset)



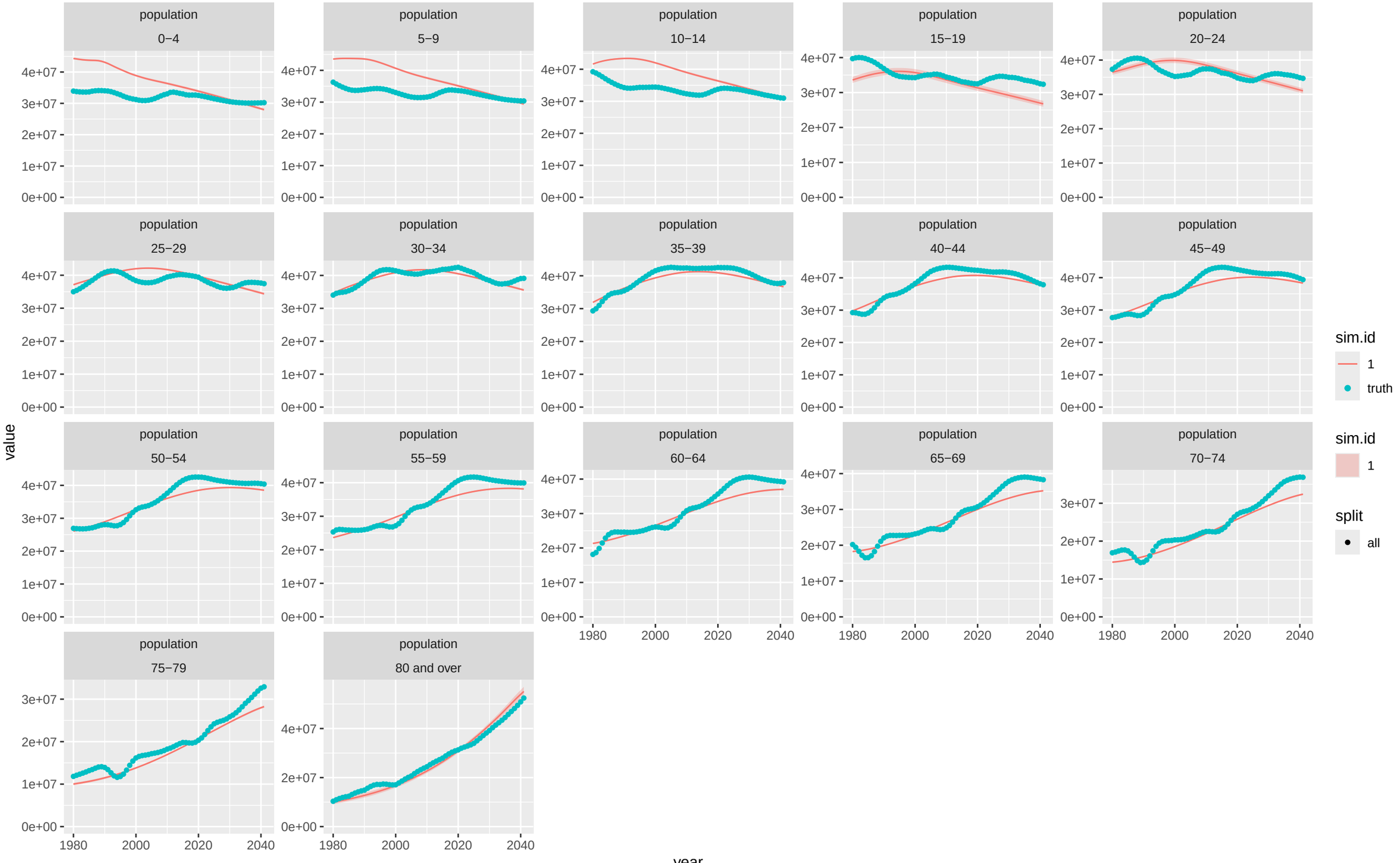
r1.high: incidence.by.age (merged Step 2 simset)



r1.high: prevalence.by.age (merged Step 2 simset)



r1.high: population.by.age (merged Step 2 simset)



r1.high: cascade (merged Step 2 simset)

awareness | engagement | suppression

value

1.00
0.75
0.50
0.25
0.00

1980 2000 2020 2040

year

sim.id
1
truth

sim.id
1

split
all

r1.high: cascade.by.age.sex (merged Step 2 simset)

awareness 0−14 male
awareness 0−14 female
awareness 15+ male
awareness 15+ female
awareness 15−24 male
awareness 15−24 female
engagement 0−14 male
engagement 0−14 female
engagement 15+ male
engagement 15+ female
engagement 15−24 male
engagement 15−24 female
suppression 0−14 male
suppression 0−14 female
suppression 15+ male
suppression 15+ female
suppression 15−24 male
suppression 15−24 female

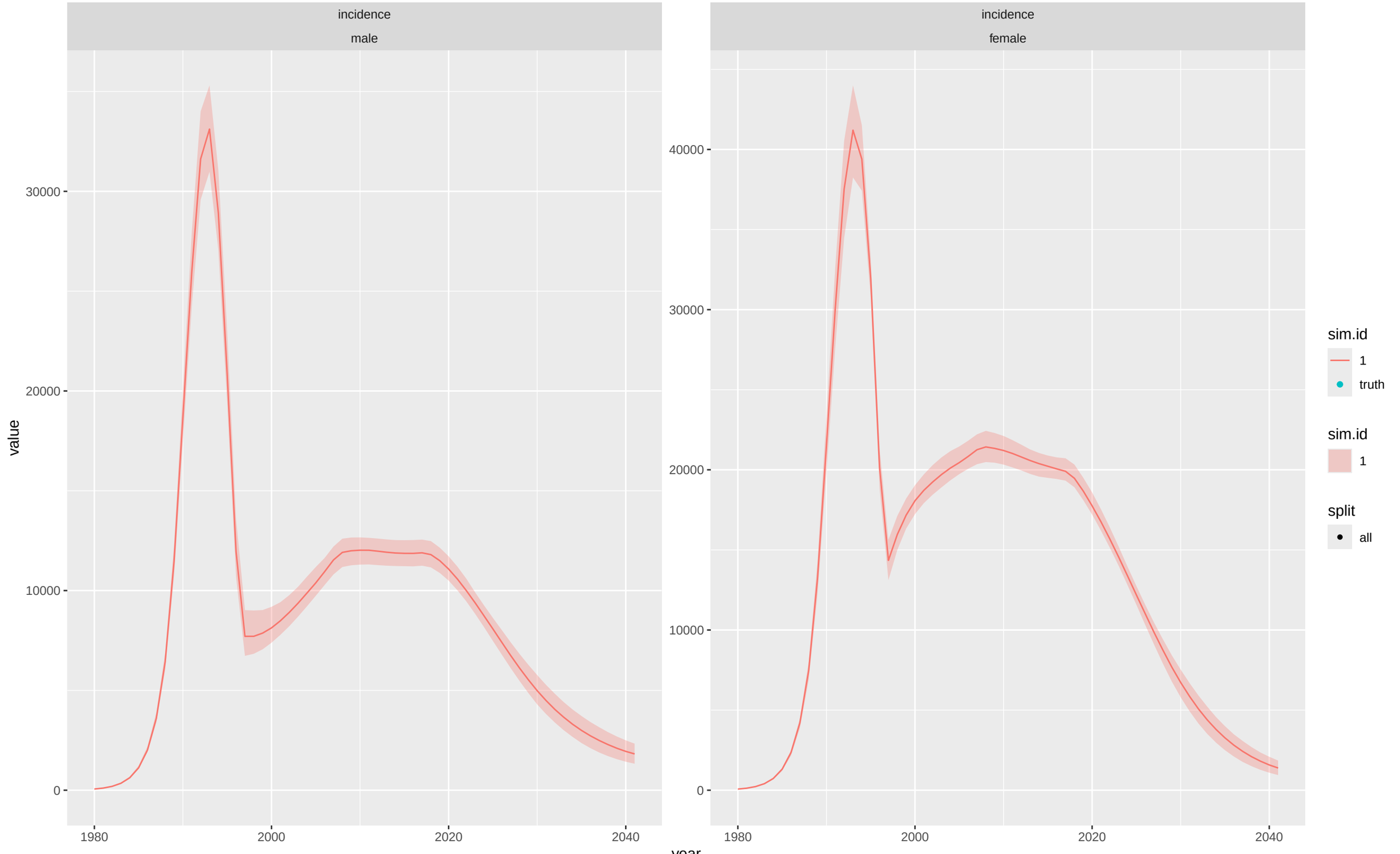
value
year

sim.id
1
truth

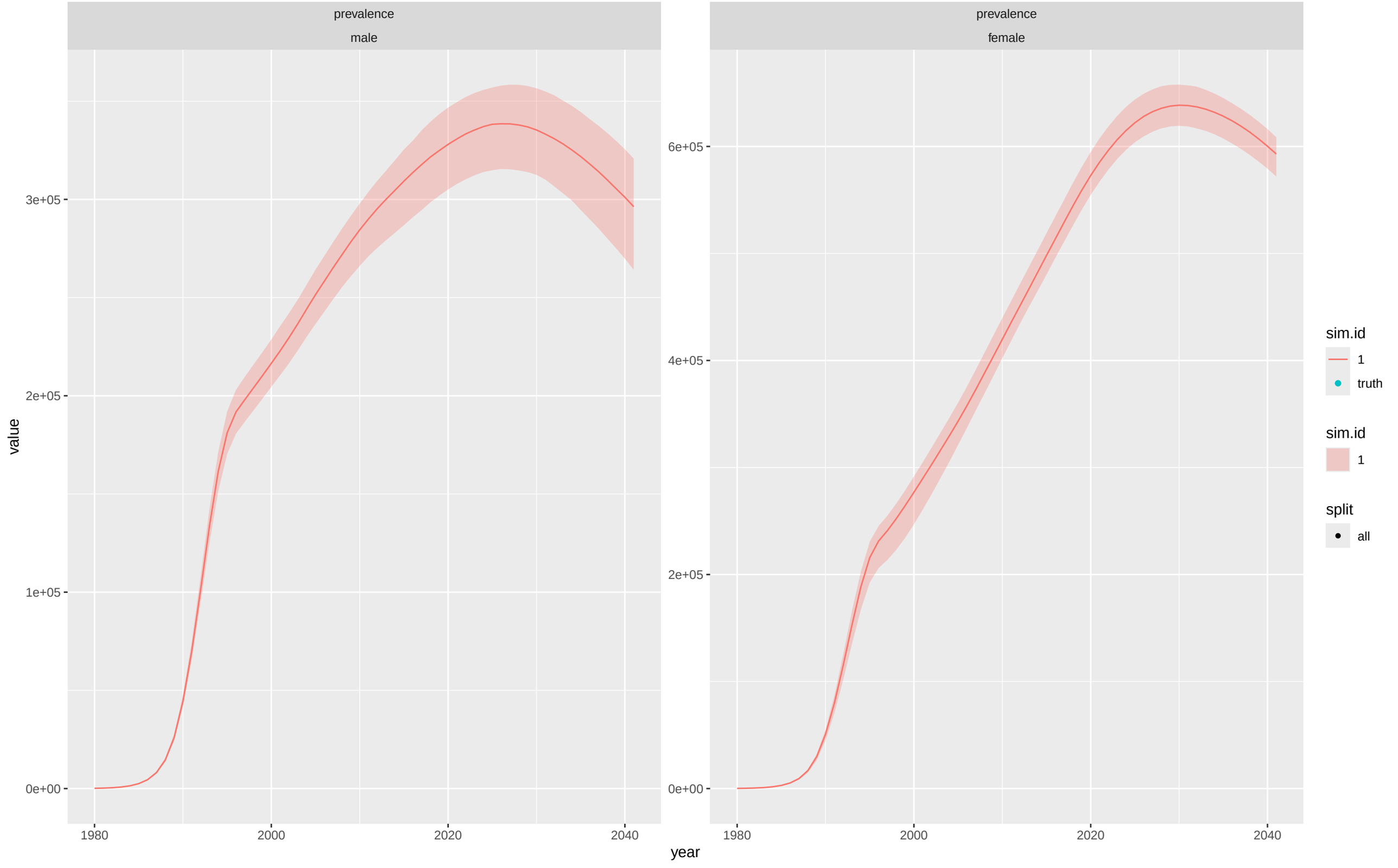
sim.id
1

split
all

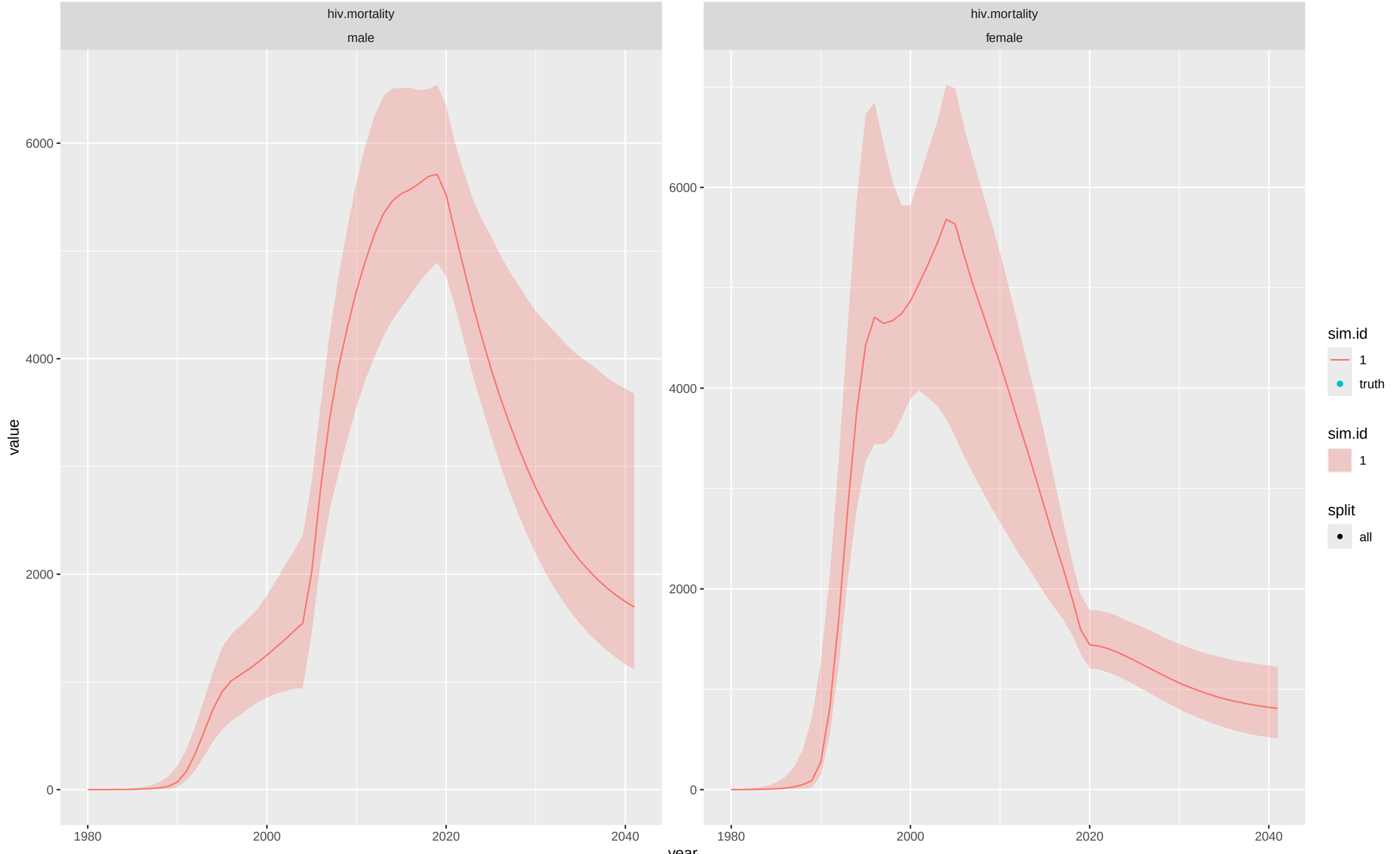
r1.high: incidence.by.sex (merged Step 2 simset)



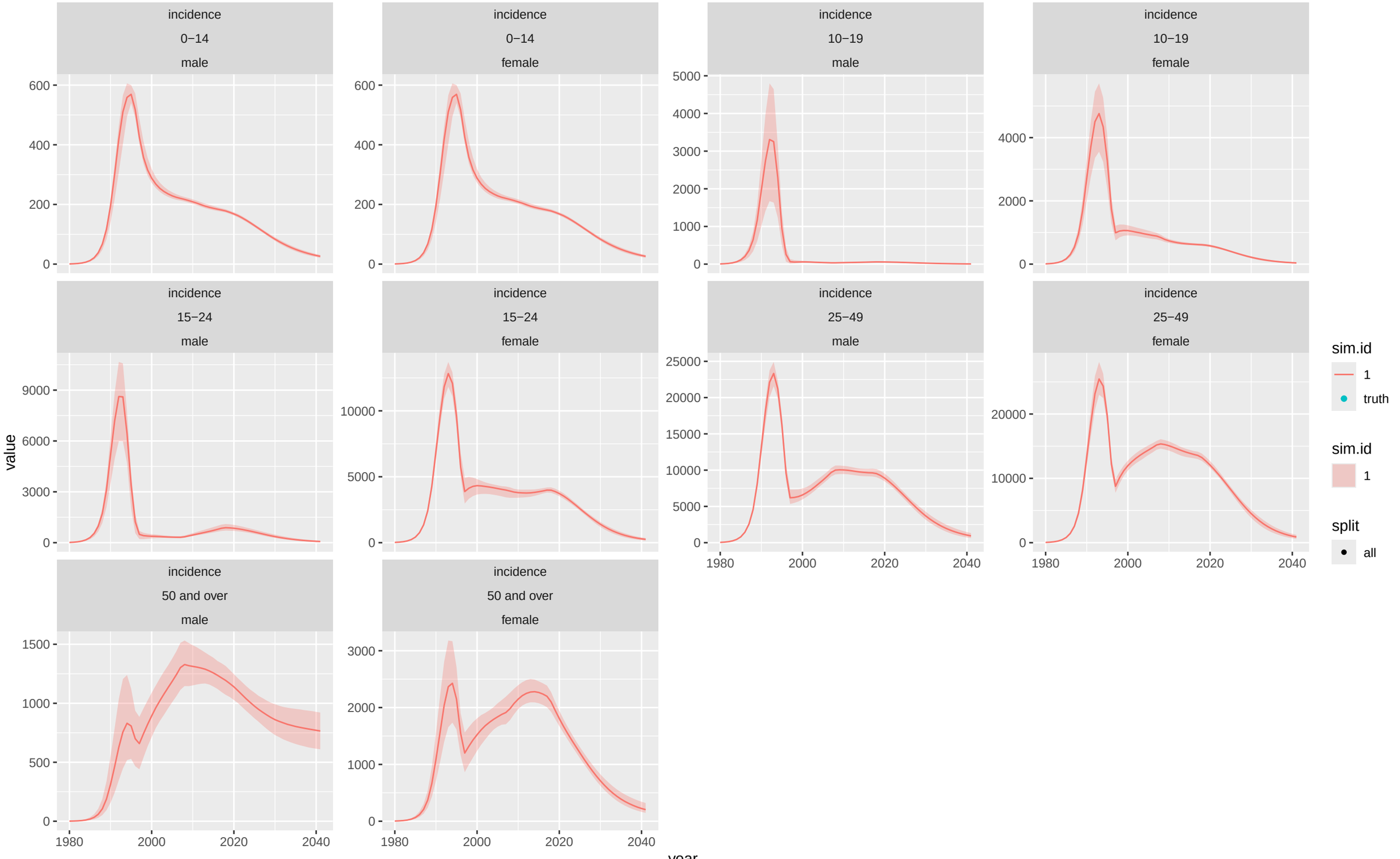
r1.high: prevalence.by.sex (merged Step 2 simset)



r1.high: hiv.mortality.by.sex (merged Step 2 simset)



r1.high: incidence.by.age.sex (merged Step 2 simset)



r1.high: prevalence.by.age.sex (merged Step 2 simset)

prevalence 0−14 male

prevalence 0−14 female

prevalence 10−19 male

prevalence 10−19 female

prevalence 15−24 male

prevalence 15−24 female

prevalence 25−49 male

prevalence 25−49 female

prevalence 50 and over male

prevalence 50 and over female

value

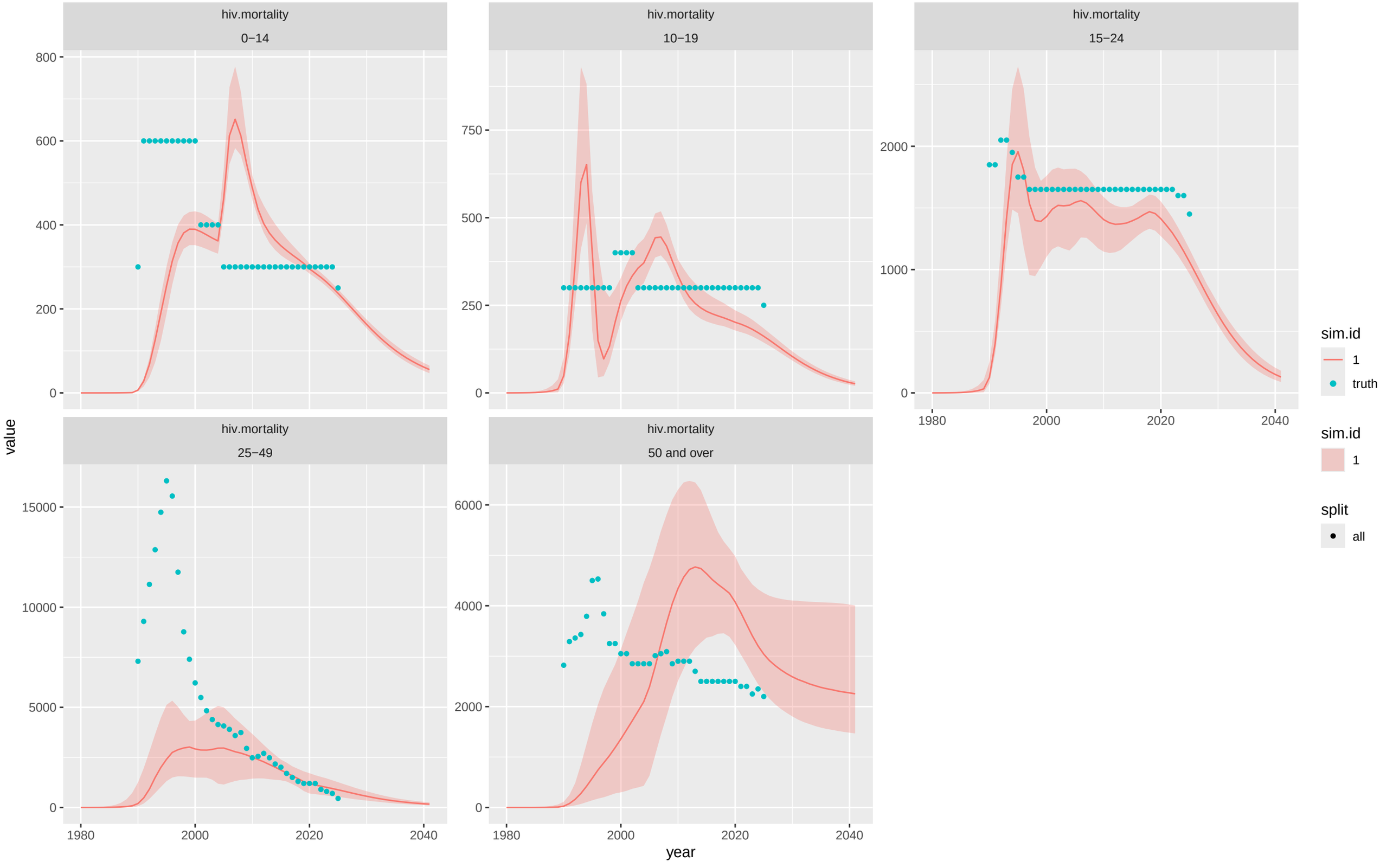
year

sim.id
1
truth

sim.id
1

split
all

r1.high: hiv.mortality.by.age (merged Step 2 simset)



r1.high: hiv.mortality.by.age.sex (merged Step 2 simset)

